

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF ACCOUNTANCY, FINANCE AND BUSINESS

ACADEMIC YEAR 2023/2024

OCTOBER EXAMINATION

BAIT3003 DATA WAREHOUSE TECHNOLOGY

MONDAY, 23 OCTOBER 2023

TIME: 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF BUSINESS (HONOURS) IN BUSINESS ANALYTICS

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

BAIT3003 DATA WAREHOUSE TECHNOLOGY

A Case Study on MR. D.I.Y

From a single store in 2005, MR. D.I.Y. has evolved to become a household name associated with everyday essentials, home improvements and more. Its purpose is to offer everyone convenience, accessibility and affordability to meet their everyday needs. Their store network grew from 900 at the close of FY2021 to 1,080 at the close of FY2022. In addition to physical stores, customers can also purchase products through its e-commerce website.

MR D.I.Y consists of the following brands:

MR D.I.Y - Always Low Prices

MR. D.I.Y. stores provide a wide variety of everyday home improvement essentials, at “Always Low Prices”, at convenient locations to meet customers’ needs.

MR. D.I.Y. Express

MR. D.I.Y. Express’ smaller store formats allow us to meet the everyday needs of customers in smaller towns and high-density locations.

MR. D.I.Y. PLUS

MR. D.I.Y. PLUS houses — MR. D.I.Y., MR.TOY and MR. DOLLAR brands in one space, offering great value and one-stop convenience, with a fun and interactive shopping experience.

MR. TOY

MR. TOY offers a diverse range of value-for-money good quality toys for children of all ages.

MR. DOLLAR

MR. DOLLAR offers food & beverages (“F&B”) and other household products at fixed prices – always RM2 or RM5 – giving customers a worry-free shopping experience.

Current state of operations

(The following discussion will use MrDIY when referring Mr D.I.Y group).

The standard **MR. D.I.Y.** stores continue to be the backbone of MrDIY's network and are the primary platform for growth and profitability.

The **MR. D.I.Y. PLUS** variant has been very successful in bringing the three core brands - **MR. D.I.Y.**, **MR. TOY** and **MR. DOLLAR** - under one roof, and will be a brand experience flagship in cities and major market centres.

MR. D.I.Y. Express, designed to meet demand for MrDIY's proven retail concept in smaller markets and high-density neighbourhoods, continues to grow from strength to strength. This retail format is to deliver on their ambition of being the store at every corner. The average MR. D.I.Y. Express store is about 4,000 sq. ft., which is consistently smaller than the average size of the standard MR. D.I.Y. store format, which is around 9,000 sq. ft..

MR. TOY made a significant comeback into profitability in FY2022 after facing pandemic-related lockdowns in FY2021, indicating that a niche for a trusted toy store brand that offers a wide range of good quality toys at affordable prices.

By comparison, **MR. DOLLAR** is still in gestation. Given prevailing consumer sentiment, there is definitely room in the market for a value retailer with a nationwide reach in this format, but with just 49 stores now, the network will need to build brand recognition, achieve scale and carry a more curated product mix to achieve more tangible success. Part of MrDIY's plans involves evolving MR. DOLLAR from a fixed price format to a “one plus” price format, which is part of their FY2023 business plan.

BAIT3003 DATA WAREHOUSE TECHNOLOGY**Performance overview**

For FY2022, 18.2% increase in revenue at RM4.0 billion, compared to RM3.4 billion for FY2021.

Total transactions grew significantly from 115 million transactions nationwide in FY2021 to more than 142 million transactions nationwide in FY2022.

The average basket size reduced from RM29.0 in FY2021 to RM27.8 in FY2022, as a result of the tightening of household wallets amidst rising costs and inflationary pressures. The impact of this on revenue was offset by the marked rise in the total number of transactions.

In terms of products, the household & furnishing categories continued to be the mainstay at 38.3%, driven by higher sales of pandemic essentials and other household products. All other categories were relatively unchanged. MrDIY's range of white label and in-house products continued to become more and more popular as customer shopping sentiment became more value-focused, and with MR. D.I.Y.-branded products increasingly known for their reliability, efficacy and accessible price points.

Product selection and merchandising mix

Data discipline is something MrDIY have always relied on to ensure they have the right products, at the right time, at the right stores. Managing their more than 20,000-SKU portfolio continues to be a key priority.

The continued success of the MrDIY's ability to deliver on their promise of "Always Low Prices" rests primarily on the continued relevance of their products, and their reputation for reliability and quality at accessible prices.

MrDIY practice a stringent approach to managing their merchandising mix. All proposed new products are evaluated based on market trends, product quality, price, and manufacturers' feedback. In addition, quarterly product reviews are held to assess product sales volumes, inventory turnover, and sales margins to determine each product's optimal shelf space.

MrDIY carry a wide range of third-party branded products, including reputable brands like Faber-Castell, Philips, Energizer and WD-40, which draw in customers and keep them returning.

MrDIY have, in recent years, successfully partnered with global manufacturers to create white-label products that carry the 'MR. D.I.Y.' or 'MR. D.I.Y. Premium' brands. These white-label products offer customers a higher price-to-quality value proposition compared to third-party branded products and are becoming increasingly popular as market demands shift to being more value-driven. In FY2022, sales of MR. D.I.Y. white-label products made up 43.5% of total sales, up from 38.7% in FY2021, with third-party branded products making up the balance. MrDIY will continue to explore introducing more white-label products to cater to market demand for products that offer exceptional value.

In FY2023, MrDIY have a store expansion target of 180 stores, which will comprise of at least 125 MR. D.I.Y. and MR. D.I.Y.PLUS stores, 35 MR. D.I.Y. Express stores, as well as 20 MR. DOLLAR and MR. TOY stores.

Adapted from: MR D.I.Y. Annual Report 2022

BAIT3003 DATA WAREHOUSE TECHNOLOGY

MrDIY stores locations



BAIT3003 DATA WAREHOUSE TECHNOLOGY

(Answer all questions, your answers should reference the case study).

Question 1

Obviously, MrDIY would have some type of Information System in place for it to do its planning and achieve strategic competitiveness.

- a) Explain why this system would most likely use at least a 3-tier system architecture. (10 marks)
- b) Suggest a suitable data warehouse architecture for MrDIY's operations. Illustrate and explain the major components of your architecture. (15 marks)

[Total: 25 marks]

Question 2

- a) There are two approaches to the design of a data warehouse:
- **top-down** (build data warehouse first)
 - **bottom-up** (build data mart first)
- Explain with relevant examples which approach is more suitable for MrDIY for it to achieve its business objectives. (10 marks)
- b) Explain with relevant examples how Mr. DIY is able to apply business intelligence with the use of a data warehouse. (Explain with reference to the **Data Warehouse Environment** and the **Analytical Environment**). (15 marks)

[Total: 25 marks]

Question 3

- a) Suggest a **Star-schema** design for MrDIY's data mart/warehouse. You may make any relevant assumptions on the attributes for the tables of the data mart/warehouse. (15 marks)
- b) Assume that there is a requirement to analyse customer orders by their state of residence. However, a customer may change their state of residence many times over their lifetime. Show how you would handle this type of slowly changing dimension. (Provide relevant examples). (10 marks)

[Total: 25 marks]

BAIT3003 DATA WAREHOUSE TECHNOLOGY**Question 4**

- a) MrDIY would like to carry out promotional sales during the major festive seasons in Malaysia. Explain how a data warehouse can maximise the information potential to assist in achieving its sales objectives. (15 marks)
- b) Backup and recovery must be properly implemented for Mr. DIY's operational database. As long as the operational database exist, we can build a data warehouse. Is there a need for backup and recovery for a data warehouse? Justify why or why not. (Provide relevant examples in your justification). (10 marks)

[Total: 25 marks]